

Neurons and neuroglia

Name _____ Date _____

Use this sheet while you work through the neurons and neuroglia walkthrough. It has 33 predictions.

1. When a step asks you to predict, write or draw your prediction in the top box, in pencil or your first color. Type it into the worksheet beside the walkthrough too. Next stays locked until you press Show me.
2. Press Show me. In the bottom box, in a second color, write what actually happened and whether you were right. If you were not, write what you would change.
3. When you finish the walkthrough, close it and do Part 2 from memory. Then reopen it and correct Part 2 in your second color. What you had to fix is what to study.

Part 1. Your predictions

PREDICTION 1 OF 33 · INFORMATION COMING IN

After a big meal, stretch receptors in your stomach wall send a signal. Which way is that signal traveling, toward the central nervous system or away from it?

My prediction, before Show me

What happened, and was I right?

PREDICTION 2 OF 33 · TWO KINDS OF TARGETS

Picking up a pencil and speeding up your heart are both commands leaving the central nervous system. Which one can you do on purpose, and which branch do you think carries each command?

My prediction, before Show me

What happened, and was I right?

PREDICTION 3 OF 33 · THE AUTONOMIC BRANCH SPLITS IN TWO

Many organs, including the heart, receive input from both autonomic branches. If one branch speeds the heart up, what would you predict the other one does to it?

My prediction, before Show me

What happened, and was I right?

PREDICTION 4 OF 33 · A NERVOUS SYSTEM IN THE GUT WALL

When food stretches a section of intestine, the smooth muscle there contracts to push the food along. This still happens in a piece of intestine whose connections to the brain and spinal cord have been cut. What does that tell you about the network in the gut wall?

My prediction, before Show me

What happened, and was I right?

PREDICTION 5 OF 33 · NEURONS SORTED BY DIRECTION

Roughly 99 percent of the neurons in your body are interneurons. Why would that be?

My prediction, before Show me

What happened, and was I right?

PREDICTION 6 OF 33 · SHAPE FOLLOWS THE JOB

You step on something sharp. That sensory signal is carried by a pseudounipolar neuron whose cell body sits off to the side, not in the signal's path. Why is that useful?

My prediction, before Show me

What happened, and was I right?

PREDICTION 7 OF 33 · EACH SHAPE HAS A HOME

Sensory fibers enter the spinal cord through the dorsal root, and efferent fibers leave through a separate ventral root. If one dorsal root is cut, what would the person lose in the area it serves, and what would they keep?

My prediction, before Show me

What happened, and was I right?

PREDICTION 8 OF 33 · WHERE THE SIGNAL COMES IN

If the branches are where signals arrive, what do you think happens to a signal that arrives at the far tip of a branch?

My prediction, before Show me

What happened, and was I right?

PREDICTION 9 OF 33 · SIGNALS THAT CHANGE SIZE

Two identical inputs arrive at the same moment, one at the far tip of a dendrite and one on the cell body right next to the decision point. Which one will have the bigger effect at the decision point?

My prediction, before Show me

What happened, and was I right?

PREDICTION 10 OF 33 · THE DECISION POINT

Why would it make sense for the decision point to sit right there, between the incoming branches and the outgoing fiber, rather than somewhere out on a branch?

My prediction, before Show me

What happened, and was I right?

PREDICTION 11 OF 33 · SAME SIZE, EVERY TIME

You press on your skin twice as hard. The signal on each sensory axon cannot get any bigger. How do you think the nervous system tells your brain the pressure went up?

My prediction, before Show me

What happened, and was I right?

PREDICTION 12 OF 33 · THE END OF THE LINE

The signal coming down the fiber is electrical. The swellings at the end do not touch the next cell directly, there is a tiny gap. What do you think has to happen for the message to cross?

My prediction, before Show me

What happened, and was I right?

PREDICTION 13 OF 33 · CALCIUM OPENS THE DOOR

A drug blocks the voltage-gated calcium channels in the axon terminal and nowhere else. An action potential arrives at the terminal. What happens?

My prediction, before Show me

What happened, and was I right?

PREDICTION 14 OF 33 · WHICH SIGNAL EACH REGION CARRIES

Before you look, sort the regions. Which ones carry signals that change size and fade, which one carries the fixed-size signal, and where does the signal stop being electrical?

My prediction, before Show me

What happened, and was I right?

PREDICTION 15 OF 33 · INSIDE OR OUTSIDE THE CNS

Which two of the six do you think work outside the brain and spinal cord? Give a reason for your guess.

My prediction, before Show me

What happened, and was I right?

PREDICTION 16 OF 33 · THE ONE THAT CONTROLS WHAT GETS IN

Every time a neuron fires, a little potassium leaks out into the narrow space around it. If astrocytes stopped clearing it, what would happen to the potassium outside, and how do you think that might change how easily the neurons fire?

My prediction, before Show me

What happened, and was I right?

PREDICTION 17 OF 33 · A WALL AROUND THE BRAIN'S BLOOD SUPPLY

With the gaps between capillary cells sealed, a substance can only reach the brain by crossing the cells themselves, through their lipid membranes. Which kinds of molecules do you predict get through easily, and which get stuck?

My prediction, before Show me

What happened, and was I right?

PREDICTION 18 OF 33 · THE ONES THAT WRAP

One oligodendrocyte wraps many fibers. One Schwann cell wraps one segment of one fiber. If a disease destroys these wrapping cells, which setting would lose more function from the same number of cells dying?

My prediction, before Show me

What happened, and was I right?

PREDICTION 19 OF 33 · THE CLEANUP CREW

Bacteria get into brain tissue. The blood–brain barrier keeps most of the body's circulating immune cells out. Which cells do you predict respond first?

My prediction, before Show me

What happened, and was I right?

PREDICTION 20 OF 33 · LINING THE FLUID SPACES

Cerebrospinal fluid is made continuously and drains out through narrow passages. If one of those passages is blocked while the fluid keeps being made, what happens upstream of the block?

My prediction, before Show me

What happened, and was I right?

PREDICTION 21 OF 33 · WHY A CUT NERVE IN YOUR ARM CAN HEAL

A person cuts a nerve in the forearm and also damages a tract in the spinal cord. Months later the forearm has recovered much of its function, but the spinal cord damage has not. Using the glial cells you have met, what might explain the difference?

My prediction, before Show me

What happened, and was I right?

PREDICTION 22 OF 33 · WHY THE GAPS MATTER

If the signal jumps from gap to gap instead of being rebuilt along every part of the fiber, what do you think that does to its speed?

My prediction, before Show me

What happened, and was I right?

PREDICTION 23 OF 33 · HOW INSULATION BUYS SPEED

Imagine a fiber with normal myelin but with the gaps placed much closer together, so there are many more of them over the same distance. Would that fiber conduct faster or slower?

My prediction, before Show me

What happened, and was I right?

PREDICTION 24 OF 33 · SODIUM CHANNELS CROWD AT THE NODES

Rebuilding the signal takes voltage-gated sodium channels. Along a myelinated fiber, where would you expect most of those channels to be?

My prediction, before Show me

What happened, and was I right?

PREDICTION 25 OF 33 · WHEN THE WRAPPING IS STRIPPED

Charge leaks out of bare fiber. If a signal now has to cross a long stripped stretch instead of jumping a short gap, what happens to it?

My prediction, before Show me

What happened, and was I right?

PREDICTION 26 OF 33 · PATIENT: NUMBNESS AND A DROPPED FOOT

Her symptoms are scattered across vision, sensation, and movement, and they come and go. Using what you just built, explain why one disease process produces such a mixed and changing picture.

My prediction, before Show me

What happened, and was I right?

PREDICTION 27 OF 33 · PATIENT: WEAKNESS CLIMBING THE LEGS

Which glial cell's myelin is under attack, and what does the near-normal response size tell you about his axons?

My prediction, before Show me

What happened, and was I right?

PREDICTION 28 OF 33 · NOTHING IS MADE OUT THERE

The terminal needs a steady supply of proteins that are only made back in the cell body. What does the cell need in order to get them there?

My prediction, before Show me

What happened, and was I right?

PREDICTION 29 OF 33 · TWO MOTOR PROTEINS

Suppose you block only the outward motor at the middle of a long axon. Where would its cargo pile up, on the cell body side of the block or on the terminal side?

My prediction, before Show me

What happened, and was I right?

PREDICTION 30 OF 33 · FAST AND SLOW

A new vesicle protein has to travel about 1,000 millimeters, one meter, from a motor neuron's cell body in the spinal cord to its terminal in the foot. At about 400 millimeters per day, roughly how long does the trip take?

My prediction, before Show me

What happened, and was I right?

PREDICTION 31 OF 33 · THE SYSTEM HAS A WEAK POINT

Rabies virus enters at a bite, gets taken up at nerve terminals in the muscle, and travels back along the fiber. Where does it end up, and why does that explain why rabies affects the brain?

My prediction, before Show me

What happened, and was I right?

PREDICTION 32 OF 33 · THE WHOLE STORY, WITH THE SCIENCE NAMES

Without scrolling back, name the regions of a neuron in the order a signal passes through them, and say what each one does and what kind of signal it carries.

My prediction, before Show me

What happened, and was I right?

PREDICTION 33 OF 33 · THE MAP, FROM MEMORY

Without scrolling back, rebuild the map in your worksheet. Start with the central nervous system, then name each division and branch, say which way information travels on each, and give one target for each efferent branch.

My prediction, before Show me

What happened, and was I right?

Part 2. Draw it from memory

The nervous system and the neuron

1. Draw the map of the nervous system

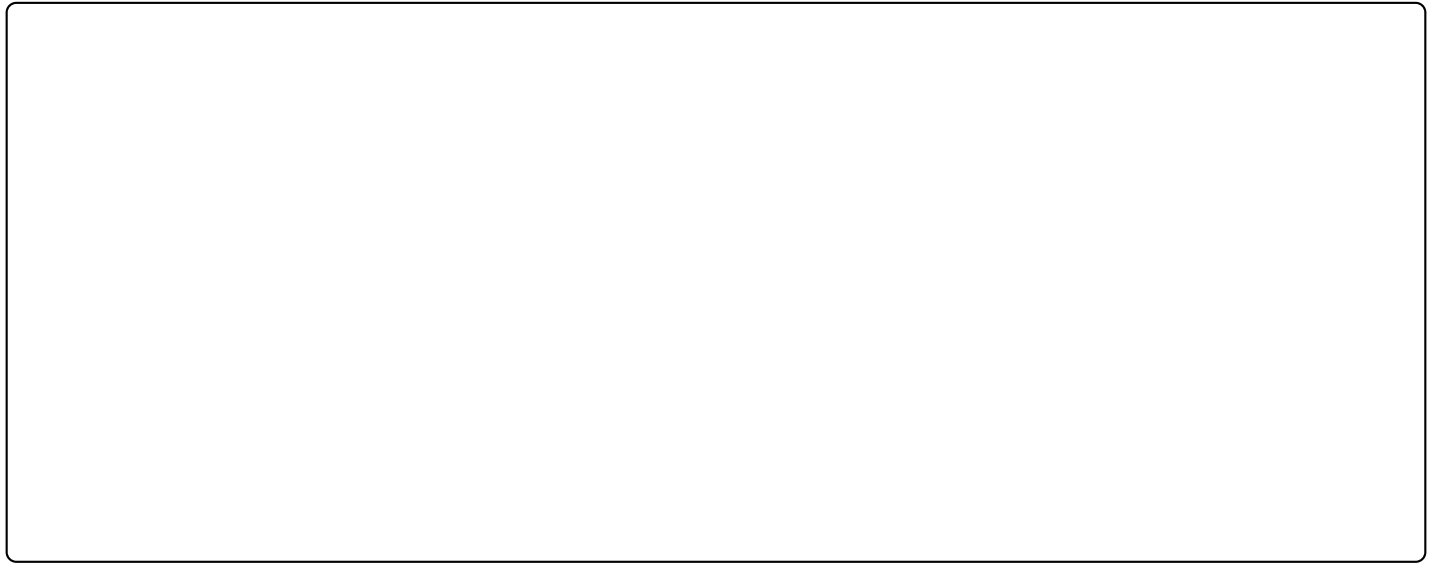
Start with the central nervous system at the top. Branch off the sensory division and the efferent division, split the efferent division into somatic motor and autonomic, split autonomic into sympathetic and parasympathetic, and add the enteric nervous system. Put an arrow on each branch showing which way information travels.

Under each efferent branch, write one target it controls.

2. Draw a multipolar neuron and label its regions

Draw it from memory. Label the dendrites, the cell body, the axon hillock and initial segment that make up the trigger zone, the axon, and the axon terminals. Then draw an arrow showing which way a signal travels through the cell.

Above each region, write the kind of signal it carries: graded, all or none, or chemical.



3. Write what happens at each stop

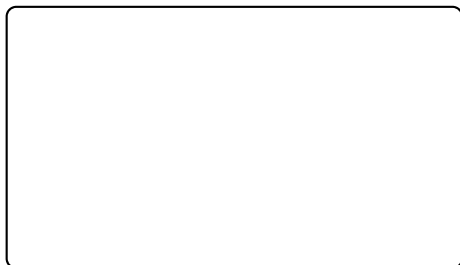
Fill in one sentence for each region, in the order a signal passes through them.



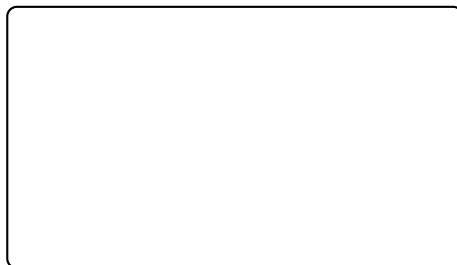
4. Three shapes

Draw a multipolar, a bipolar, and a pseudounipolar neuron. Under each one, write where in the body you would find it and which branch of your map it belongs on.

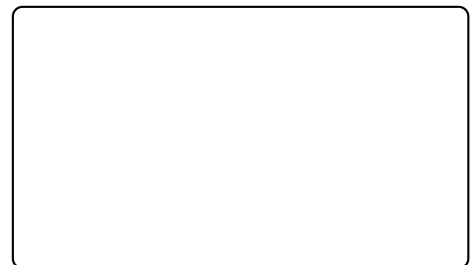
Multipolar



Bipolar



Pseudounipolar



The support cells, myelin, and axonal transport

1. Draw each glial cell and write its job

One cell per box, drawn from memory. Under each drawing write what that cell does, and whether it lives in the central nervous system or outside it.

Astrocyte

Oligodendrocyte

Microglia

Ependymal cell

Schwann cell

Satellite cell

2. Myelin, and what happens without it

On the left, draw a myelinated axon. Show the covered stretches, the bare gaps, where the sodium channels are crowded, and how the signal moves along it. On the right, draw the same axon with two stretches of myelin destroyed, and show what happens to the signal.

Healthy

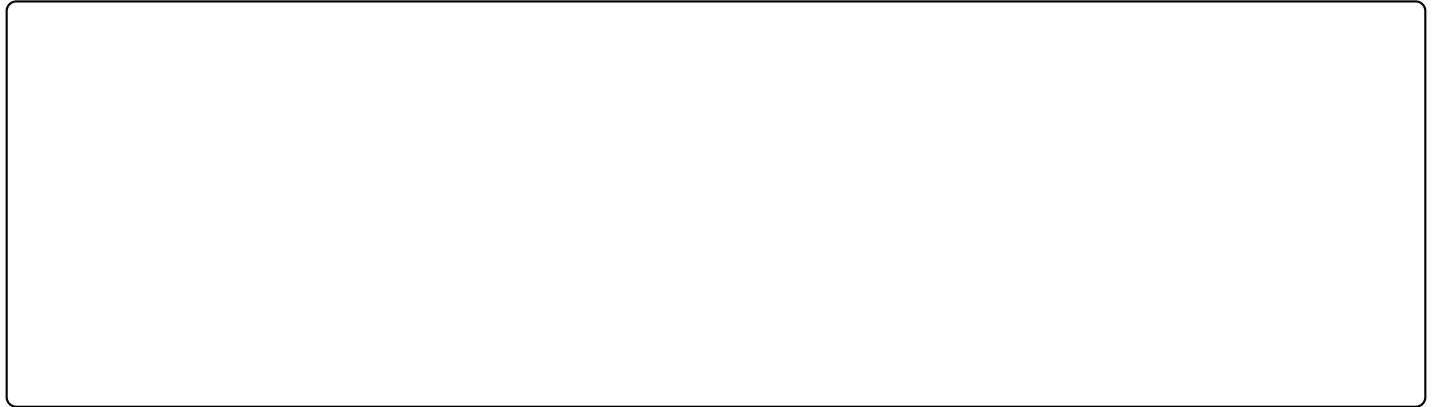
Myelin stripped

Then answer: why do the symptoms of multiple sclerosis show up in so many unrelated places?

3. Traffic on the axon

Draw a long neuron with the cell body on the left and the terminals on the right, with a microtubule track running down the axon. Draw a package moving outward and one moving back, and label the motor protein that carries each. Add the slow stream as a third, much slower package.

Beside each package, write what it carries and roughly how fast it moves.



Then answer: how does a virus that enters at a nerve terminal reach the cell body, and which motor protein carries it?

